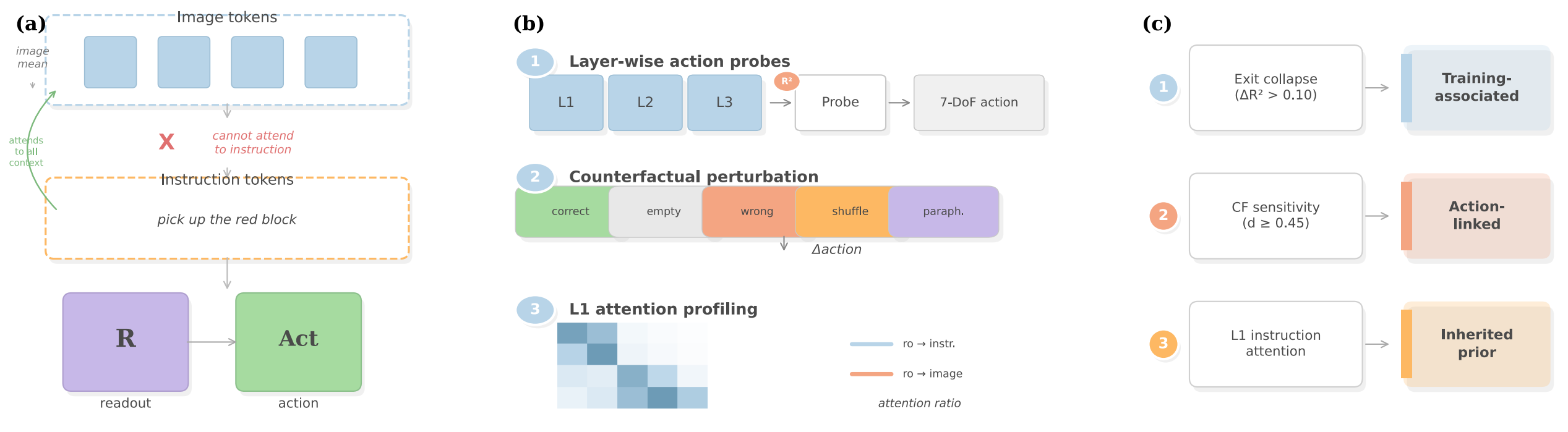

(a)
Image tokens
image mean
attends to all context
X
cannot attend to instruction
Instruction tokens
pick up the red block
R
Act
readout
action
(b)
1
Layer-wise action probes
L1
L2
L3
R²
Probe
7-DoF action
2
Counterfactual perturbation
correct
empty
wrong
shuffle
paraph.
Δaction
3
L1 attention profiling
ro → instr.
ro → image
attention ratio
(c)
1
Exit collapse
(ΔR² > 0.10)
Training-associated
2
CF sensitivity
(d ≥ 0.45)
Action-linked
3
L1 instruction attention
Inherited prior